

Data Engineering

Introduction

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What is Data Engineering?

❖ Definition:

- The development, implementation, and maintenance of systems that take raw data and produce high-quality, consistent information for downstream use.

❖ The Core Mission:

- To take data from "Source" to "Value".

❖ The Intersection:

- It sits at the crossroads of Software Engineering, Data Architecture, Security, and Operations (DataOps).

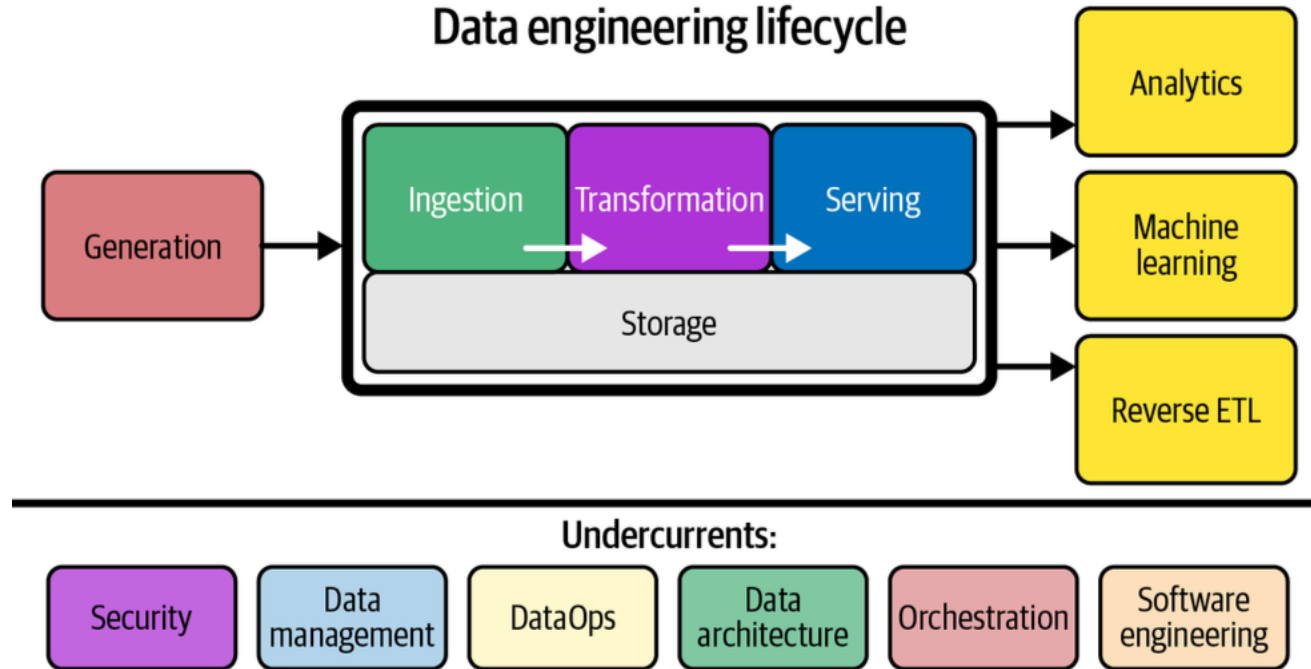
The Data Engineering Lifecycle

❖ The Core Mental Model

- Instead of focusing on tools (Spark vs. Kafka), focus on stages.

- ❖ 1. **Generation**: Source systems (Apps, IoT, APIs) create data.
- ❖ 2. **Storage**: Persisting data in Lakes or Warehouses.
- ❖ 3. **Ingestion**: Moving data from source to storage.
- ❖ 4. **Transformation**: converting raw data into usable models.
- ❖ 5. **Serving**: Delivering data to consumers (Dashboards, ML Models, Reverse ETL).

The Data Engineering Lifecycle



The Data Engineering Lifecycle

- ❖ The "Undercurrents"
 - Concepts that apply across every stage of the lifecycle.
- ❖ **Security:**
 - Protecting data and systems, which includes access control and privacy management.
- ❖ **Data Management:**
 - This encompasses data governance, data quality, and making data discoverable (e.g., using data catalogs).
- ❖ **DataOps:**
 - The application of DevOps principles to data, focusing on automation, observability, and "agile" delivery of data products.
- ❖ **Data Architecture:**
 - The overall design of systems, balancing trade-offs like cost, scalability, and simplicity to meet business goals.
- ❖ **Orchestration:**
 - Managing the dependencies and workflow execution between different tools and stages of the pipeline.
- ❖ **Software Engineering:**
 - The fundamental technical skills required to build these systems, including coding (Python/SQL), version control, and testing

The Evolution of the Field

❖ From Warehouse to Big Data

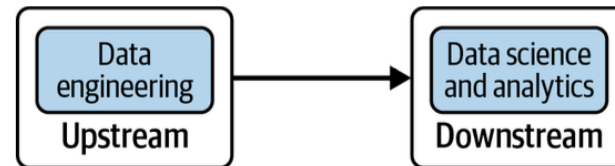
- 1980s-2000s (The Warehouse Era): SQL-centric. Oracle/Teradata. ETL Developers managing monolithic, on-premise systems.
- 2000s-2015 (The Big Data Era): The "3 Vs" (Volume, Velocity, Variety). Hadoop & MapReduce. "Big Data Engineers" managing complex clusters of commodity hardware.
- **Key Shift:** Moving from vertical scaling (bigger machines) to horizontal scaling (more machines).

❖ The Modern Era (2020s)

- The Lifecycle Era: Shift from "Managing Infrastructure" to "Managing Data."
- Simplification: Cloud-native tools abstract away the hardware.
- The **Modern Data Stack**: Modular, best-of-breed tools acting like "LEGO bricks."

Data Engineering vs. Data Science

❖ Upstream vs. Downstream



❖ **Upstream** (Data Engineering):

- Focuses on the inputs. Building the pipelines, infrastructure, and ensuring data quality/availability.

❖ **Downstream** (Data Science/Analytics):

- Focuses on the outputs. Building models, dashboards, and deriving business insights.

❖ The Relationship:

- Data Engineering enables Data Science. Without solid engineering, data scientists spend 80% of their time cleaning data.

The Hierarchy of Data Needs

- ❖ Base (Infrastructure/Pipelines):
 - The "Data Engineering" layer. Reliable flow, storage, and collection.
- ❖ Middle (Analytics/Metrics):
 - The "doing business" layer.
- ❖ Top (AI/Deep Learning):
 - High visibility, but requires a solid base.
- ❖ Lesson: **Building the Foundation first**
 - You cannot effectively do AI (Top) if you haven't solved Ingestion/Storage (Bottom).

THE DATA SCIENCE HIERARCHY OF NEEDS

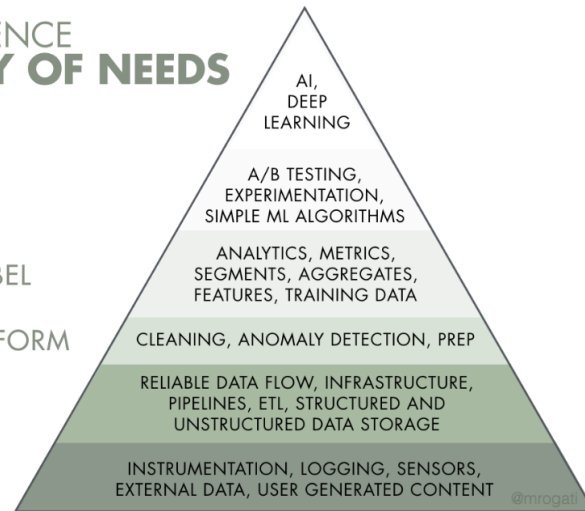
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

COLLECT



Data Maturity Models – Where is the organization?

❖ Stage 1:

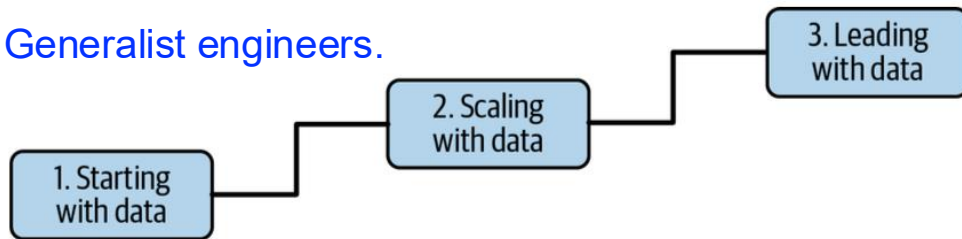
- **Starting with Data:** Ad-hoc requests. Generalist engineers.
- Goal: "Quick Wins" to prove value.
- Risk: Tech debt.

❖ Stage 2:

- **Scaling with Data:** Formal practices. Specialized roles.
- Goal: Scalable architecture.
- Risk: Over-engineering.

❖ Stage 3:

- **Leading with Data:** Self-service analytics. Automated governance. Data is a product.
- Risk: Complacency.



The Continuum of Roles

❖ Type A vs. Type B Data Engineers

❖ Type A (**Abstraction**):

- Focuses on using off-the-shelf tools to solve business problems quickly. "Don't reinvent the wheel." (Common in Stage 1 & 2 companies).

❖ Type B (**Build**):

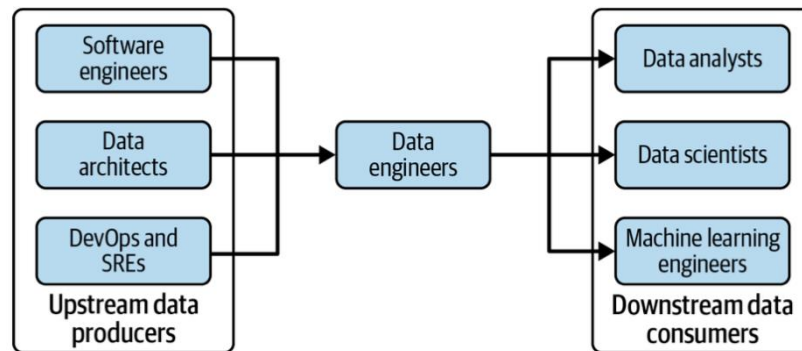
- Focuses on building custom data tools and infrastructure to handle massive scale or unique problems. (Common in Stage 3 or Tech Giants).

❖ Trend:

- The industry is shifting toward Type A as managed tools become better and cheaper.

Who do Data Engineers Work With?

- ❖ Upstream (Producers):
 - Software Engineers, Architects. (DEs must communicate requirements to them).
- ❖ Downstream (Consumers):
 - Data Scientists, Analysts. (DEs must understand their needs).
- ❖ Leadership:
 - CTOs/CDOs. (DEs provide the "Art of the Possible" for business strategy).



Key Skills for the Modern Engineer

- ❖ Software Engineering:
 - Coding (Python/Java), Version Control (Git), Testing.
- ❖ SQL:
 - The "Lingua Franca" of data. Essential for transformation and analysis.
- ❖ System Architecture:
 - Understanding trade-offs (Batch vs. Stream, SQL vs. NoSQL).
- ❖ Communication:
 - The ability to translate between "Business Needs" and "Technical Implementation."

Summary: The Modern Landscape

❖ Foundation:

- Data Engineering builds the platform that makes AI and Analytics possible.

❖ Lifecycle:

- Think in stages (Ingest -> Transform -> Serve), not just tools.

❖ Evolution:

- The role has moved from "Infrastructure and Data Plumber" to "Data Lifecycle Architect."

❖ Goal:

- To deliver high-quality, reliable data to the business.

References

- ❖ Reis & Housley, Fundamentals of Data Engineering, O'Reilly, 2022.
 - <https://learning.oreilly.com/library/view/fundamentals-of-data/9781098108298/>

